Chapter 8. Optimisation: Convergence, Momentum and Adaptive Methods

Lectures: Week 4 Wed, Week 5 Mon (Taylor expansion and descent lemma, SGD, momentum, Nesterov, AdaGrad, RMSProp, Adam)

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1 Gradient Descent and Taylor's Theorem

We minimise $f : \mathbb{R}^d \to \mathbb{R}$; the iterate $\mathbf{x}_t$ is the whole parameter vector. GD is $\mathbf{x}_{t+1} = \mathbf{x}_t - \eta \nabla f(\mathbf{x}_t)$, and $-\nabla f$ is the steepest-descent direction since $\nabla f^\top \mathbf{v} \geq -\|\nabla f\|$ for unit $\mathbf{v}$ (Cauchy–Schwarz).

Theorem: Taylor along a segment

With $\mathbf{d} = \mathbf{y} - \mathbf{x}$ and $h(t) = f(\mathbf{x} + t\mathbf{d})$,

$$f(\mathbf{y}) - f(\mathbf{x}) = \int_0^1 \nabla f(\mathbf{x}+t\mathbf{d})^\top \mathbf{d}\, dt, \qquad f(\mathbf{y}) = f(\mathbf{x}) + \nabla f(\mathbf{x})^\top \mathbf{d} + \int_0^1 (1-t)\, \mathbf{d}^\top \nabla^2 f(\mathbf{x}+t\mathbf{d})\mathbf{d}\, dt.$$

Proof. The first is $h(1) - h(0) = \int_0^1 h'$. For the second, $h(1) = h(0) + h'(0) + \int_0^1 \int_0^t h''(s)\, ds\, dt$ and exchanging the order over $0 \leq s \leq t \leq 1$ gives $\int_0^1 (1-s) h''(s)\, ds$. ■

2 Smoothness and the Descent Lemma

Definition: L-smoothness

f is L-smooth if $\|\nabla f(\mathbf{x}) - \nabla f(\mathbf{y})\| \leq L\|\mathbf{x} - \mathbf{y}\|$ for all $\mathbf{x}, \mathbf{y}$. For $f \in C^2$ this is equivalent to $-L\mathbf{I} \preceq \nabla^2 f \preceq L\mathbf{I}$ (all Hessian eigenvalues in $[-L, L]$).

Theorem: descent lemma

If f is L-smooth, $f(\mathbf{y}) \leq f(\mathbf{x}) + \nabla f(\mathbf{x})^\top (\mathbf{y} - \mathbf{x}) + \frac{L}{2}\|\mathbf{y} - \mathbf{x}\|^2$.

Proof. Subtract $\nabla f(\mathbf{x})^\top \mathbf{d} = \int_0^1 \nabla f(\mathbf{x})^\top \mathbf{d}\, dt$ from the first Taylor formula, then use Cauchy–Schwarz and smoothness:

$$f(\mathbf{y}) - f(\mathbf{x}) - \nabla f(\mathbf{x})^\top \mathbf{d} = \int_0^1 (\nabla f(\mathbf{x} + t\mathbf{d}) - \nabla f(\mathbf{x}))^\top \mathbf{d}\, dt \leq \int_0^1 Lt\|\mathbf{d}\|^2 dt = \frac{L}{2}\|\mathbf{d}\|^2.$$

■

Theorem: GD on smooth functions

GD satisfies $f(\mathbf{x}_{t+1}) \leq f(\mathbf{x}_t) - \eta(1 - \frac{L\eta}{2})\|\nabla f(\mathbf{x}_t)\|^2$, so it decreases f when $0 < \eta < 2/L$. With $\eta = 1/L$ and $f^\star = \inf f > -\infty$,

$$\min_{0 \leq t < T} \|\nabla f(\mathbf{x}_t)\|^2 \leq \frac{1}{T} \sum_{t<T} \|\nabla f(\mathbf{x}_t)\|^2 \leq \frac{2L\,(f(\mathbf{x}_0) - f^\star)}{T}.$$

Proof. Put $\mathbf{y} = \mathbf{x}_t - \eta \nabla f(\mathbf{x}_t)$ in the descent lemma. With $\eta = 1/L$ the coefficient is $1/(2L)$; summing over t telescopes the left side to $f(\mathbf{x}_0) - f(\mathbf{x}_T) \leq f(\mathbf{x}_0) - f^\star$. ■

This is a stationarity guarantee: for nonconvex f a small gradient may sit at a saddle point or a local minimum. For $f(x) = x^2/2$ ($L = 1$), $x_{t+1} = (1-\eta)x_t$ converges iff $0 < \eta < 2$; for $f = \frac{1}{2}\mathbf{x}^\top H \mathbf{x}$ each eigen-direction contracts by $1 - \eta\lambda_i$, so a small eigenvalue (poor conditioning) means slow progress while $\eta < 2/\lambda_{\max}$ is required for stability.

3 Stochastic Gradient Descent

SGD uses $\mathbf{x}_{t+1} = \mathbf{x}_t - \eta \mathbf{g}_t$ with $\mathbb{E}_t \mathbf{g}_t = \nabla f(\mathbf{x}_t)$, where $\mathbb{E}_t$ conditions on the history up to $\mathbf{x}_t$.

Theorem: second moment is not variance

$\mathbb{E}_t\|\mathbf{g}_t\|^2 = \|\nabla f(\mathbf{x}_t)\|^2 + \mathbb{E}_t\|\mathbf{g}_t - \nabla f(\mathbf{x}_t)\|^2.$

Proof. Expand $\mathbf{g}_t = \nabla f + (\mathbf{g}_t - \nabla f)$; the cross term has conditional mean zero. ■

Theorem: $T^{-1/2}$ rate for SGD

If f is L-smooth, $\mathbb{E}_t\|\mathbf{g}_t\|^2 \le G$ and $\Delta = f(\mathbf{x}_0) - f^\star$, then for a constant step

$$\frac{1}{T}\sum_{t<T}\mathbb{E}\|\nabla f(\mathbf{x}_t)\|^2 \le \frac{\Delta}{\eta T} + \frac{L\eta G}{2}, \qquad \text{and with } \eta = \sqrt{\tfrac{2\Delta}{LGT}}: \quad \le \sqrt{\frac{2L\Delta G}{T}}.$$

Proof. Descent lemma: $f(\mathbf{x}_{t+1}) \le f(\mathbf{x}_t) - \eta\nabla f(\mathbf{x}_t)^\top\mathbf{g}_t + \frac{L\eta^2}{2}\|\mathbf{g}_t\|^2$. Conditional expectation gives $\mathbb{E}_t f(\mathbf{x}_{t+1}) \le f(\mathbf{x}_t) - \eta\|\nabla f(\mathbf{x}_t)\|^2 + \frac{L\eta^2}{2}G$. Take total expectation (tower property), sum over $t < T$ (telescoping), use $\mathbb{E}f(\mathbf{x}_T) \ge f^\star$ and divide by ηT. Finally $\frac{a}{\eta} + b\eta$ is minimised at $\eta = \sqrt{a/b}$ with value $2\sqrt{ab}$. ■

The noise term forces $\eta \sim T^{-1/2}$, slower than GD's T^{-1}. Since mini-batches of size B divide the noise variance by B, larger batches reduce G and permit larger steps.

4 Momentum

4.1 Heavy-ball momentum

Definition: momentum (lecture convention)

$\mathbf{v}_{t+1} = \beta\mathbf{v}_t - \eta\nabla f(\mathbf{x}_t)$, $\mathbf{x}_{t+1} = \mathbf{x}_t + \mathbf{v}_{t+1}$, $\mathbf{v}_0 = \mathbf{0}$, $0 \le \beta < 1$.

Unrolling, $\mathbf{v}_{t+1} = -\eta\sum_{k=0}^{t}\beta^{t-k}\nabla f(\mathbf{x}_k)$: an exponentially weighted sum of past gradients (a velocity). For a constant gradient $\mathbf{g}$, $\|\mathbf{v}_t\| \to \eta\|\mathbf{g}\|/(1-\beta)$, an effective step $1/(1-\beta)$ times larger ($10\times$ for $\beta = 0.9$). Persistent directions accumulate; oscillating directions cancel, which reduces zig-zagging in narrow valleys. Physically the gradient acts as a force and $1-\beta$ as friction: in the buffer form $\mathbf{m}_{t+1} = \mathbf{g}_t + \beta\mathbf{m}_t$ one has $\mathbf{m}_{t+1} - \mathbf{m}_t = \mathbf{g}_t - (1-\beta)\mathbf{m}_t$.

Theorem: momentum on a quadratic

For $f(x) = \frac{h}{2}x^2$, $x_{t+1} = (1+\beta-\eta h)x_t - \beta x_{t-1}$; the iteration converges iff both roots of $r^2 - (1+\beta-\eta h)r + \beta = 0$ lie in the open unit disk, which for $0 \le \beta < 1$ holds iff $0 < \eta h < 2(1+\beta)$. Complex roots (modulus $\sqrt{\beta}$) mean damped oscillation.

Proof. Substitute $\mathbf{v}_{t+1} = x_{t+1} - x_t$ and $\mathbf{v}_t = x_t - x_{t-1}$ into the velocity update. For the root condition use the Jury/Schur criterion for $r^2 + a_1 r + a_0$: $|a_0| < 1$, $1 + a_1 + a_0 > 0$, $1 - a_1 + a_0 > 0$, which here read $\beta < 1$, $\eta h > 0$, $\eta h < 2(1+\beta)$. ■

4.2 Nesterov momentum

Nesterov evaluates the gradient at the look-ahead point:

$$\mathbf{y}_t = \mathbf{x}_t + \beta\mathbf{v}_t, \qquad \mathbf{v}_{t+1} = \beta\mathbf{v}_t - \eta\nabla f(\mathbf{y}_t), \qquad \mathbf{x}_{t+1} = \mathbf{x}_t + \mathbf{v}_{t+1}.$$

Example: lecture numbers

$f = x^2/2$, $x_0 = 1$, $\eta = 0.1$, $\beta = 0.9$. Both methods give $v_1 = -0.1$, $x_1 = 0.9$. Momentum: $v_2 = 0.9(-0.1) - 0.1(0.9) = -0.18$, $x_2 = 0.72$. Nesterov: $y_1 = 0.81$, $v_2 = -0.09 - 0.081 = -0.171$, $x_2 = 0.729$: it measures the slope where it is about to go and brakes earlier (not necessarily closer to the optimum at every step).

Theorem: two forms of Nesterov (2021 Q6)

Form I: $\psi^{k+1} = \theta^k - \alpha g^k$, $\theta^{k+1} = \psi^{k+1} + \beta(\psi^{k+1} - \psi^k)$, $\psi^0 = \theta^0$. Form II (PyTorch `nesterov=True`): $m^{k+1} = g^k + \beta m^k$, $v^{k+1} = \beta m^{k+1} + g^k$, $\theta^{k+1} = \theta^k - \alpha v^{k+1}$, $m^0 = 0$. Here g^k is evaluated at θ^k. For the same θ^0 and $g^0, g^1, \ldots$ both produce the same $\theta^1, \theta^2, \ldots$

Proof. *Both satisfy one recursion.* Form I with $\psi^{k+1} = \theta^k - \alpha g^k$ and $\psi^k = \theta^{k-1} - \alpha g^{k-1}$ gives, for $k \geq 1$,

$$\theta^{k+1} = \theta^k - \alpha g^k + \beta(\theta^k - \theta^{k-1}) - \alpha\beta(g^k - g^{k-1}).$$

Form II: $\theta^{k+1} = \theta^k - \alpha g^k - \beta(\alpha g^k + \alpha\beta m^k)$, and from the previous step $\alpha\beta m^k = \alpha v^k - \alpha g^{k-1} = \theta^{k-1} - \theta^k - \alpha g^{k-1}$; substituting gives the same recursion. Both start with $\theta^1 = \theta^0 - \alpha(1+\beta)g^0$. A second-order recursion with the same two initial values has the same solution (induction). *Alternative.* Define $\psi^k := \theta^k + \alpha\beta m^k$ in Form II. Then $\psi^0 = \theta^0$, $\psi^{k+1} = \theta^k - \alpha g^k$ and $\psi^{k+1} - \psi^k = -\alpha m^{k+1}$, from which Form I's second equation follows. ■

Interpretation: ψ is the actual position (the slide's x_t), θ the look-ahead point $x_t + \beta v_t$ at which gradients are taken and which PyTorch stores as the parameter; m is the velocity of ψ scaled by $-1/\alpha$. Both converge to the same limit since $\theta^k - \psi^k = \beta(\psi^k - \psi^{k-1}) \to 0$.

5 Adaptive Methods

All operations below are coordinatewise.

Definition: AdaGrad, RMSProp, Adam

AdaGrad: $\mathbf{r}_t = \mathbf{r}_{t-1} + \mathbf{g}_t \odot \mathbf{g}_t$, $\mathbf{x}_{t+1} = \mathbf{x}_t - \eta\, \mathbf{g}_t \oslash (\sqrt{\mathbf{r}_t} + \epsilon)$.
RMSProp: $\mathbf{r}_t = \rho \mathbf{r}_{t-1} + (1-\rho)\mathbf{g}_t \odot \mathbf{g}_t$, same update.
Adam: $\mathbf{m}_t = \beta_1 \mathbf{m}_{t-1} + (1-\beta_1)\mathbf{g}_t$, $\mathbf{v}_t = \beta_2 \mathbf{v}_{t-1} + (1-\beta_2)\mathbf{g}_t \odot \mathbf{g}_t$, $\hat{\mathbf{m}}_t = \mathbf{m}_t/(1-\beta_1^t)$, $\hat{\mathbf{v}}_t = \mathbf{v}_t/(1-\beta_2^t)$, $\mathbf{x}_t = \mathbf{x}_{t-1} - \eta\, \hat{\mathbf{m}}_t \oslash (\sqrt{\hat{\mathbf{v}}_t} + \epsilon)$. Typical $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\eta = 10^{-3}$.

Remark: what each piece does

AdaGrad's accumulator grows forever, so its effective rate $\eta/\sqrt{r_t}$ shrinks (for a constant gradient g the step is $\eta/\sqrt{t}$). RMSProp forgets old gradients; for constant $g = 2$, $\rho = 0.9$, $\eta = 0.1$, $r_1 = 0.4$, $r_2 = 0.76$, steps $0.316, 0.229$, tending to 0.1. Adam = momentum's numerator (direction) + RMSProp's denominator $\sqrt{\mathbf{v}}$ (per-coordinate scale) + bias correction. (Nesterov plus Adam is NAdam.)

5.1 Exponential moving averages and bias correction (2024 Q1)

Theorem: EMA of iid gradients

If $g_0, g_1, \dots$ are iid with mean μ and variance s^2, and $m_{k+1} = \beta m_k + (1-\beta) g_k$, $m_0 = 0$, then

$$\mathbb{E} m_k = (1-\beta^k)\mu, \qquad \operatorname{Var} m_k = s^2 \frac{1-\beta}{1+\beta}(1-\beta^{2k}),$$

with limits μ and $s^2 \frac{1-\beta}{1+\beta}$. The corrected $\hat{m}_k = m_k/(1-\beta^k)$ is unbiased with variance $s^2 \frac{(1-\beta)(1+\beta^k)}{(1+\beta)(1-\beta^k)}$ (equal to s^2 at $k=1$).

Proof. m_k depends only on $g_0, \dots, g_{k-1}$, so it is independent of g_k. Mean by induction: $\beta(1-\beta^k)\mu + (1-\beta)\mu = (1-\beta^{k+1})\mu$. Variance: $\operatorname{Var} m_{k+1} = \beta^2 \operatorname{Var} m_k + (1-\beta)^2 s^2$; writing $(1-\beta)^2 = \frac{1-\beta}{1+\beta}(1-\beta^2)$, the induction step is $\beta^2(1-\beta^{2k}) + (1-\beta^2) = 1 - \beta^{2k+2}$. Correction: divide by $(1-\beta^k)^2$ and use $1-\beta^{2k} = (1-\beta^k)(1+\beta^k)$. ■

Without correction Adam's first step is $\frac{1-\beta_1}{\sqrt{1-\beta_2}} \operatorname{sign}(g_1)\eta \approx 3.16\,\eta$; with correction $\hat{m}_1 = g_1$, $\hat{v}_1 = g_1^2$ and the step is exactly $\eta \operatorname{sign}(g_1)$. The correction removes the zero-initialisation bias at the price of a larger early variance.

5.2 Special cases and invariances (2022 Q1)

Theorem: Adam contains RMSProp and signSGD

For Adam without bias correction: $\beta_1 = 0$ gives RMSProp with $\rho = \beta_2$; $\beta_1 = \beta_2 = 0$, $\epsilon = 0$ give signSGD $\theta_{k+1} = \theta_k - \eta \operatorname{sign}(g_k)$ (if no coordinate of g_k is 0). With $\epsilon = 0$, Adam's iterates are unchanged when the loss is multiplied by $c > 0$, while SGD's step scales by c.

Proof. $\beta_1 = 0$ makes $m_{k+1} = g_k$. With $\beta_2 = 0$ also, $\sqrt{v_{k+1}} = |g_k|$. Scaling $g \mapsto cg$ scales m by c and $\sqrt{v}$ by c. ■

5.3 Weight decay is not ℓ_2 regularisation (2024 Q7)

For plain SGD, ℓ_2 regularisation $\theta - \eta(g + \lambda\theta) = (1-\eta\lambda)\theta - \eta g$ is the same as weight decay. With momentum it is not:

Theorem: no matching exists

SGD with momentum $v_{k+1} = g_k + \beta v_k$, $\theta_{k+1} = \theta_k - \alpha v_{k+1}$ and $\alpha, \beta > 0$. Weight decay: $\theta_{k+1} = \theta_k - \alpha v_{k+1} - \mu\theta_k$. ℓ_2: $v_{k+1} = (g_k + \lambda\theta_k) + \beta v_k$. No choice of $\mu, \lambda > 0$ makes the two methods equivalent.

Proof. Take $g_k \equiv 0$, $v_0 = 0$, $\theta_0 \neq 0$. Weight decay: $v_k = 0$ and $\theta_k = (1-\mu)^k \theta_0$. ℓ_2: $v_1 = \lambda\theta_0$, $\theta_1 = (1-\alpha\lambda)\theta_0$, $v_2 = \lambda(1 - \alpha\lambda + \beta)\theta_0$, $\theta_2 = [(1-\alpha\lambda)^2 - \alpha\beta\lambda]\theta_0$. Matching θ_1 forces $\mu = \alpha\lambda$, and then matching θ_2 forces $\alpha\beta\lambda = 0$, a contradiction. (Indeed the ℓ_2 sequence is geometric for no ratio.) ■

In Adam the ℓ_2 gradient $\lambda\theta$ is divided by its own RMS, so every weight moves by about η regardless of its size ($\theta_1 = \theta_0 - \eta \operatorname{sign}(\theta_0)$ when $g \equiv 0$), losing proportional shrinkage; AdamW applies the decay $(1-\eta\lambda)\theta$ outside the adaptive scaling.